



Binomial Distribution Topic Test 1

Solution

Total Marks: 30

Time: 50 minutes

- 1 The weights in a population are normally distributed with a mean of 80 kg and a standard deviation of 8 kg. Use the empirical rule to find the approximate probability that a randomly selected person has a weight under 64 kg? 2

64 kg is 2 standard deviation less than 80 kg ✓
So the answer is 0.025

- 2 (i) Records show that 64% of students at a school travelled to and from school by bus. Samples of 100 students at the school are taken to determine the proportion who travel to and from school by bus. Show that the distribution of such sample proportions has mean 0.64 and standard deviation 0.048. 2

(i) For students in the sample, the number of students traveling to and from school by bus is a random variable X with the Binomial distribution $B(100, 0.64)$. Hence the sample proportion $\frac{X}{100}$ has mean $\mu = 0.64$ since $E\left(\frac{X}{n}\right) = \frac{np}{n} = p$ ✓
and standard deviation $\sigma = \sqrt{\frac{0.64 \times (1-0.64)}{100}} = 0.048$ ✓

- (ii) Use the table below of $P(Z < z)$, where Z has a standard normal distribution, to estimate the probability that a sample of 100 students will contain at least 58 and at most 64 students who travel to and from school by bus. 2

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177

(ii) $P(0.58 \leq \frac{X}{100} \leq 0.64)$
Z scores for 0.64 = $\frac{0.64 - 0.64}{0.048} = 0$
Z scores for 0.58 = $\frac{0.58 - 0.64}{0.048} = -1.25$ ✓
 $P(-1.25 \leq Z \leq 0)$
= $P(Z \leq 0) - P(Z \leq -1.25)$
= $0.5 - 0.1038$
= 0.3962 ✓

- 3 Shares in a company X are particularly volatile. On any given trading day the probability that the share price will increase by \$1 is $\frac{3}{5}$ and the probability that the share price will fall by \$1 is $\frac{2}{5}$. If the share price is now \$6 find the probability that it will be less than \$6 after 5 trading days. 2

$$\begin{aligned} \text{Prob}(< \$6 \text{ after 5 days}) &= \text{Prob}(\text{Majority of movements are down}) \\ &= \text{Prob}(\text{number of up movements} \leq 2) \\ &= \binom{5}{0} \left(\frac{2}{5}\right)^0 \left(\frac{3}{5}\right)^5 + \binom{5}{1} \left(\frac{2}{5}\right)^1 \left(\frac{3}{5}\right)^4 + \binom{5}{2} \left(\frac{2}{5}\right)^2 \left(\frac{3}{5}\right)^3 = \frac{992}{3125} \end{aligned}$$

- 4 On a roulette wheel, there are 18 red numbers, 18 black numbers, and 1 green numbers. A ball is dropped onto the spinning wheel and lands on one of the numbers randomly. Each result is independent. A gambler bets that the ball land on any of the black numbers.

- (i) If the gambler makes the same bet five times, let the random variable Y be the number of times the gambler wins. Write the distribution of Y in standard form, and give its mean and variance. 2

This is a binomial random variable
with $p = \frac{18}{37}$ and $n = 5$.
 $Y \sim \text{Bin}\left(5, \frac{18}{37}\right)$ ✓
 $\mu = 5 \times \frac{18}{37} = \frac{90}{37} \approx 2.43$ ✓
 $\sigma^2 = 5 \times \frac{18}{37} \times \frac{19}{37} = \frac{1710}{1369} \approx 1.25$ ✓

- (ii) If the gambler makes the same bet five times, what is the probability he will win more times than he loses? Give your answer correct to three decimal places. 2

This is a binomial probability
and we are looking for $P(W \geq 3)$
where W is the number of wins
in 5 bets
 $P(W \geq 3) = P(W=3) + P(W=4) + P(W=5)$
 $= \binom{5}{3} \left(\frac{18}{37}\right)^3 \left(\frac{19}{37}\right)^2 + \binom{5}{4} \left(\frac{18}{37}\right)^4 \left(\frac{19}{37}\right)^1 + \binom{5}{5} \left(\frac{18}{37}\right)^5 \left(\frac{19}{37}\right)^0$
 $= 0.475$ ✓

- 5 Weather observations in the town of Dampville have established that the probability of rain on any given day is 0.8. Observations are made for 100 consecutive days.

Let X be the random variable representing the number of rainy days.

- (i) Find the expected value $E(X)$. $E(X) = np = 100 \times 0.8 = 80$ days 1
- (ii) Find the standard deviation of X . $\sigma = \sqrt{npq} = \sqrt{100 \times 0.8 \times 0.2} = 4$ 1
- (iii) By considering normal distribution find the approximate probability that $76 \leq x \leq 84$. 1

This represents one standard deviation from the mean so
 $P(76 \leq X \leq 84) \approx 0.68$ ✓

- 6 In a dice game five dice are rolled simultaneously. One point is scored for each six showing face-up on the dice. Let X represent the score.

- (i) Explain why X can be considered as a Bernoulli variable. 1
- (ii) Find the expected value and variance of X . 1
- (iii) Find $P(X > 3)$, correct to four decimal places. 2

X has a binomial distribution. Its shape is positively skewed.

$$E(X) = np = 5 \times \frac{1}{6} = \frac{5}{6} \quad (\text{see Reference Sheet})$$

$$\text{Var}(X) = np(1-p) = 5 \times \frac{1}{6} \times \frac{5}{6} = \frac{25}{36} \quad (\text{see Reference Sheet})$$

$$P(X > 3) = P(X = 4 \text{ or } X = 5) = P(X = 4) + P(X = 5)$$

$$= \binom{5}{4} \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^1 + \binom{5}{5} \left(\frac{1}{6}\right)^5 \left(\frac{5}{6}\right)^0 \quad \text{or} \quad \binom{5}{4} \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^1 + \left(\frac{1}{6}\right)^5 \quad (\text{see Reference Sheet})$$

$$= 0.0033 \text{ (4 d.p.)}$$

7 In a certain school, 23% of Year 12 students study Mathematics Extension 1.

(i) If X represents the number of students who study Mathematics Extension 1, describe the skewness of the binomial distribution for $P(X = x)$. You must give reasons for your answer. 1

(ii) The Principal meets with a random sample of 60 Year 12 students, to discuss their Year 12 HSC courses. What is the probability that more than 30% of the students that meet with the Principal study Mathematics Extension 1? (You may assume that the sample of students is approximately normally distributed, and make reference to this extract below from a table of z scores). 4

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
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(i) Since $p = 0.23 < 0.5$, where p is the probability of a student studying Mathematics Extension 1, then the distribution is skewed to the right, i.e. positive skewness.

(ii) $\mu = p = 0.23$

$\sigma^2 = \frac{p(1-p)}{n}$ $\sigma^2 = \frac{0.23 \times 0.77}{60}$ $\sigma^2 = 0.0029516\ldots$ $\sigma = 0.054329\ldots$	$z = \frac{x - \mu}{\sigma}$ $z = \frac{0.3 - 0.23}{0.054329}$ $z = 1.2884\ldots$ $z = 1.29$
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$$P(z < 1.29) = 0.9015$$

$$\text{Therefore, } P(z > 1.29) = 1 - 0.9015 = 0.0985$$

8 The probability that a certain drug cures a disease is 0.8. The drug is administered to 10 000 patients.

Let X be the binomial random variable representing the number of patients cured.

(i) Find $E(X)$. (i) $E(X) = np = 10000 \times 0.8 = 8000$ 1

(ii) Show that X has a standard deviation of 40. (ii) $Var(X) = np(1-p)$ 1

$$= 10000 \times 0.8 \times 0.2 = 1600$$

$$\sigma = \sqrt{1600} = 40$$

(iii) Use a normal approximation to find the probability that X is less than 8040. 1

$$(iii) P(X < 8040) = P(Z < 1) = 0.5 + 0.34 = 0.84$$

9 An online furniture store claims that 90% of all orders are shipped within 54 hours of a customer placing an order through the store's website. Ty ordered a total of 150 various pieces of furniture from the store for his company.

(i) According to the store's claim, 135 pieces of furniture should be delivered within 54 hours of Ty's order being placed through the store's website. What is the probability that this delivery will occur? Give your answer to three significant figures. 1

(ii) Show that the expected value and the standard deviation of the sampling proportion are 0.9 and 0.0245 respectively. 2

$$(i) P(x = 135) = \binom{150}{135} 0.1^{15} \times 0.9^{135}$$

$$= 0.10797\ldots \approx 0.108$$

$$(ii) E(\hat{p}) = p = 0.9$$

$$Var(\hat{p}) = \frac{0.9(1-0.9)}{150} = \frac{3}{5000}$$

$$SD = \sqrt{\frac{3}{5000}} = 0.0245$$



Binomial Distribution Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 A multiple-choice test contains ten questions. Each question has four choices for the correct answer. Only one of the choices is correct.

- (i). What is the probability of getting 90% with random guessing? 1

$$P(\text{correct}) = \frac{1}{4} \quad P(\text{incorrect}) = \frac{3}{4} \quad n = 10$$

$$P(90\%) = P(\text{exactly 9 correct}) = {}^{10}C_9 \left(\frac{3}{4}\right)^1 \left(\frac{1}{4}\right)^9 = \frac{15}{524288}$$

- (ii). What is the probability of getting at most 90% with random guessing? 2

$$P(\text{At most 90\%}) = 1 - P(100\%) = 1 - {}^{10}C_{10} \left(\frac{1}{4}\right)^{10} = 1 - \left(\frac{1}{4}\right)^{10} = 1 - \frac{1}{1048576}$$

- 2 The probability that the 7:30 a.m. train arrives on time is 0.85. Find an expression for the probability that the train is on time at least five days during one week. 2

$$\begin{aligned} P(\geq 5 \text{ out of } 7) \\ &= {}^7C_5 (0.85)^5 (0.15)^2 + {}^7C_6 (0.85)^6 (0.15)^1 + {}^7C_7 (0.85)^7 \\ &= 21(0.85)^5 (0.15)^2 + 7(0.85)^6 (0.15)^1 + (0.85)^7 \approx 0.9262 \end{aligned}$$

- 3 The random variable X is distributed binomially such that $X \sim \text{Bin}(n, 0.2)$. If the mean of X is double the size of the standard deviation of X , what is the size of n ? 2

$$\begin{aligned} E(X) &= 2 \times \sigma \\ n \times 0.2 &= 2 \times \sqrt{n \times 0.2 \times 0.8} \\ 0.1n &= \sqrt{0.16n} \\ 0.01n^2 - 0.16n &= 0 \\ \therefore n &= 16 \end{aligned}$$

- 4 From a standard deck of cards, a card is drawn 20 times, with replacement. What is the probability that a red card is drawn exactly 15 times? 1

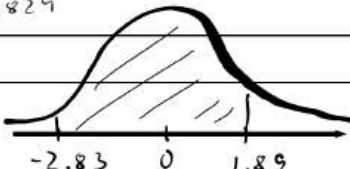
$$\begin{aligned} \text{Bin}(20, 0.5) &= {}^{20}C_{15} \times \left(\frac{1}{2}\right)^{15} \times \left(\frac{1}{2}\right)^5 \\ &= {}^{20}C_{15} \times \left(\frac{1}{2}\right)^{20} \\ &= \frac{{}^{20}C_{15}}{2^{20}} \end{aligned}$$

- 5 In a lollypop factory the material for making each lollypop is sent through a machine. The time, X seconds, taken to produce a lollypop by the machine is a binomial distribution where it can be shown that $P(X \leq 3) = \frac{5}{29}$. A random sample of 12 lollypops is chosen. What is the probability that exactly 5 of these 12 lollypop took 3 or less seconds to produce? 1

$$P(X \leq 3) = \frac{5}{29} \Rightarrow p \quad q = 1 - p = \frac{24}{29}$$

- 6 In the first half of 2022, 66% of all passengers departing from Sydney airport travelled internationally. A survey of 500 randomly selected people on the airport was conducted. Assuming that the sampling distribution of proportions $\hat{p}$ is approximately normally distributed, estimate the probability that the percentage of international travellers in the sample lies between 60% and 70%.

3

$$\begin{aligned}
 &\text{Population : } p = 0.66 \quad q = 1 - p = 0.34 \\
 &\text{[3] Sample : } \hat{p}, \mu_{\hat{p}} = p = 0.66, n = 500 \\
 &P(0.6 \leq \hat{p} \leq 0.7) \\
 &= P\left(\frac{0.6 - 0.66}{\sqrt{\frac{0.66 \times 0.34}{500}}} \leq z \leq \frac{0.7 - 0.66}{\sqrt{\frac{0.66 \times 0.34}{500}}}\right) \\
 &= P(-2.8322... \leq z \leq 1.8813...) \\
 &= P(-2.83 \leq z \leq 1.89) \\
 &= P(z \leq 1.89) - P(z \leq -2.83) \\
 &= P(z \leq 1.89) - P(z \geq 2.83) \\
 &= P(z \leq 1.89) - 1 + P(z \leq 2.83) \quad \text{Table} \\
 &= 0.97062 - 1 + 0.99767 \\
 &= 0.96829
 \end{aligned}$$


- 7 Tomatoes are considered to be either determinate or indeterminate.

Mario buys tomatoes from his local store where the tomatoes are twice as likely to be sourced from Farm A than Farm B.

It is known that 60% of Farm A's tomatoes are determinate while 70% of Farm B's tomatoes are determinate. Mario cannot tell the difference between these tomatoes from their appearance.

- (i). Show that the probability that a randomly selected tomato is determinate is $\frac{19}{30}$.

1

$$\begin{aligned}
 P(\text{determinate}) &= P(\text{Farm A and determinate}) + P(\text{Farm B and determinate}) \\
 &= \frac{2}{3} \times 60\% + \frac{1}{3} \times 70\% = \frac{19}{30}
 \end{aligned}$$

- (ii). Mario buys ten tomatoes. Find the probability, correct to two decimal places, that no more than either of these are determinate.

2

Let X be the random variable representing the number of determinate tomatoes.

$$\begin{aligned}
 P(X \leq 8) &= 1 - P(X = 9) - P(X = 10) \\
 &= 1 - {}^{10}C_9 \left(\frac{19}{30}\right)^9 \left(\frac{11}{30}\right)^1 - {}^{10}C_{10} \left(\frac{19}{30}\right)^{10} \left(\frac{11}{30}\right)^0 = 0.93
 \end{aligned}$$

- (iii). Mario buys 33 tomatoes to make a batch of tomato paste and knows from his grandmother that at least 55% of the tomatoes should be determinate to achieve a fine paste. By using a normal approximation to the sample proportion, determine the approximate probability that the tomato paste produced will be considered fine.

3

Let $\hat{p}$ be the random variable representing the sample proportion of determinate tomatoes.

$$E(\hat{p}) = p = \frac{19}{30}$$

$$\text{Standard deviation } (\hat{p}) = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{\frac{19}{30} \times \frac{11}{30}}{33}} = \sqrt{\frac{19}{2700}}$$

We need to find

$$P(\hat{p} \geq 0.55) = P\left(Z \geq \frac{0.55 - \frac{19}{30}}{\sqrt{\frac{19}{2700}}}\right) \\ = P(Z \geq -1)$$

$$= 0.34 + 0.5, \text{ using z-score empirical rules} = 0.84$$

- 8 In an experiment, a pair of dice are rolled 80 times. A success is recorded if the sum of the dice is 10 or more.

- (i). What is the mean of this binomial distribution? Give your answer to one decimal place.

3

$$\begin{array}{c|ccc} & \dots & 4 & 5 & 6 \\ \hline \vdots & & & & \\ 4 & & & 10 & 11 \\ 3 & & 10 & 11 & 12 \\ 2 & & 10 & 11 & 12 \end{array}$$

$$\text{let } P(\text{10 or more}) = \frac{6}{36} \quad n=80 \\ = \frac{1}{6}$$

$$\text{Let } X = \text{number of rolls} \geq 10$$

$$X \sim \text{Bin}\left(80, \frac{1}{6}\right)$$

$$E(X) = np \\ = 80 \cdot \frac{1}{6} \\ = 13.3 \quad (1 \text{ dp})$$

- (ii). What is the standard deviation? Give your answer to one decimal place.

1

$$\text{Var}(X) = np(1-p) \\ \sigma^2 = 80 \cdot \frac{1}{6} \cdot \frac{5}{6} \\ = 11.1 \\ \therefore \sigma = 3.3 \quad (1 \text{ dp})$$

- 9 A phone manufacturer knows that 4% of the mobile phones it produces will be defective. Phones are shipped in boxes of 20 phones. Boxes are shipped on pallets containing 200 boxes. Calculate the probability that:

- (i). A box of mobile phones contains exactly 3 defective phones. 1

Let p be the probability that a phone is defective $\Rightarrow p = 0.04$

Let X be the number of defective phones
 $X \sim \text{Bin}(20, 0.04)$

$$P(X=3) = {}^{20}C_3 (0.04)^3 (0.96)^{17} \\ = 0.0364499$$

- (ii). A box of mobile phones contains at least 1 defective phone. 1

$$P(X \geq 1) = 1 - P(X=0) \\ = 1 - {}^{20}C_0 (0.04)^0 (0.96)^{20} \\ = 0.557998$$

- (iii). A pallet contains between 90 and 95 (inclusive) boxes with at least 1 defective phone. 3

Use the probability table provided.

$$n = 200 \quad p = 0.558$$

$$\mu = np \\ = 111.6$$

$$\sigma = \sqrt{np(1-p)} \\ = 7.02$$

$$P(90 \leq X \leq 95) = P(3.08 \leq Z \leq -2.36) \\ = P(Z \leq 3.08) - P(Z \leq 2.36) \\ = 0.9990 - 0.9909 \\ = 0.0081$$

10 Records show that 20% of students at a school play a musical instrument. A sample of 100 students at the school is to be taken to determine the proportion who play a musical instrument.

- (i). Find the mean and standard deviation of the distribution of such a sample proportion. 2

$$(e) \quad p = \frac{20}{100} = 0.2, \quad q = 0.8, \quad n = 100$$

$$(i) \quad \text{Mean: } E(\hat{p}) = p = 0.2 \quad \sigma^2 = 0.0016$$

$$\text{std dev: } \text{Var}(\hat{p}) = \frac{pq}{n} \quad \sigma = 0.04$$

- (ii). Use the following extract of the table of values of $P(Z \leq z)$, where Z has a standard normal distribution, to estimate the probability that a sample of 100 students at the school contains at least 18 and at most 26 who play a musical instrument. 3

STANDARD NORMAL DISTRIBUTION TABLE

Entries represent $P(Z \leq z)$. The value of z to the first decimal is given in the left column. The second decimal is given in the top row.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767

$$(i) \quad \text{for } X = 18, \quad z = \frac{0.18 - 0.2}{0.04} = -0.5$$

$$P(Z > -0.5) = 1 - P(Z < -0.5) = 1 - 0.3085 = 0.6915$$

$$\text{for } X = 26, \quad z = \frac{0.26 - 0.2}{0.04} = 1.5$$

$$P(Z < 1.5) = 0.9332$$

$$\text{Then, } P(-0.5 < Z < 1.5) = 0.9332 - 0.3085 = 0.6247$$

